Omair Shafiq

PhD in Computational Structural & Earthquake Engineering

+60-1127588189 • omair.shafiq1998@gmail.com
linkedin.com/in/omair-shafiq/ • researchgate.net/profile/Omair-Shafiq
G.E. / BEM Malaysia (G2263729A) • P.E. / PEC (CIVIL/54467)

Professional Summary

PhD researcher and Design Engineer (P.E.) with 6+ years of combined research and industry experience in structural and earthquake engineering. Author of **10 journal publications** spanning Q1–Q3 venues, with four papers currently under review in *Structures* (Q1) and *MethodsX* (Q1). Originator of the **EIDA Framework** — an end-to-end automated pipeline for seismic fragility assessment integrating C#, MATLAB, and the ETABS API across thousands of parametric nonlinear analyses. Combines deep computational expertise with hands-on structural design and transportation engineering practice.

Education

2024–2027 **Doctor of Philosophy — Structural Engineering**, *Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA)*, Malaysia
- Thesis: AI-integrated seismic fragility assessment and performance-based structural design.
- Supervisor: Dr. Saffuan Bin Wan Ahmad. Research outputs: 5 journal papers submitted/under review (4 in Q1 venues).

2020–2024 **Master of Science — Structural Engineering**, *University of Engineering and Technology (UET)*, Peshawar, Pakistan
- Thesis: Development of Fragility Curves for Single-Span Small Bridges in Peshawar (Conference Paper Published).

2015–2019 **Bachelor of Science — Civil Engineering**, *University of Engineering and Technology (UET)*, Peshawar, Pakistan
- Thesis: Achieving High-Strength Concrete in a Challenging Environment (Published — GRIN Verlag, 2019).

Doctoral Research Contributions

EIDA Framework (Core PhD Output)
- Designed and deployed the **Automated Extended-IDA (EIDA) Framework** — a high-throughput C# + MATLAB pipeline enabling thousands of parametric nonlinear time-history analyses via the ETABS API, with automated topology extraction, stochastic material assignment, Hunt-and-Fill ground motion scaling to global collapse (DS1–DS5), and real-time fragility curve generation.
- Programmed a C# API controller for 3D FE mesh generation, material constitutive modelling, and result extraction; developed a Direct Text Injection Protocol to modify nonlinear hinge properties and boundary conditions in real time.
- Integrated Soil–Structure Interaction via Gazetas spring formulations with lumped plasticity modelling across thousands of RC frame configurations.

Meta-Regression & Machine Learning
- Fitted GLME models (REML) per damage state to quantify marginal parameter influences on fragility (DS1–DS5), revealing SSI-induced reversals in member-proportioning effects.

- Developed a two-stage transparent ML framework: black-box benchmarking to establish predictive ceilings, then sparse regression (LASSO/Elastic Net) on three dimensionless macro-parameters to yield closed-form equations — under review in Structures (Q1).

Key Research & Technical Strengths

Computational Structural Dynamics & Automation

- Expert in parametric nonlinear time-history analysis, fragility curve generation, and seismic vulnerability assessment at scale using C#, MATLAB, and the ETABS API.
- Proficient in probabilistic seismic demand modelling (PSDM), incremental dynamic analysis (IDA), and limit-state identification for RC frame structures.
- Applied machine learning (XGBoost, ANN, Sparse Regression) to seismic fragility derivation; developed transparent closed-form dimensionless inputs fragility equations from ML outputs.

Earthquake Engineering

- Developed seismic fragility curves for single-span slab bridges in Peshawar, Pakistan.
- Generated artificial and spectrally matched acceleration time-history records for Pakistan seismic hazard zones.

Structural Design

- Developed *Conc* — software for beam/slab capacity and column interaction diagram calculation (C#, WPF, .NET).
- Validated and redesigned steel frame designs for truck loading in Australia.
- Designed and drafted the Women's Business Development Centre for KPCIP in Kohat.

Road Asset Management & Transportation

- Extensive hands-on experience with RAM equipment: Profilometer and Falling Weight Deflectometer (FWD).
- Developed *Flexpave* — flexible pavement design software using AASHTO 2007 (C#, WPF, .NET).
- Experienced in HDM-4 for road policy guidance; carried out road safety audits on N-55 (Indus Highway) and PRIP roads.

Professional Experience

2022–2026 **Design Engineer**, *CEC Minconsult*, 4 years

- Supervised transportation projects individually valued at over $1 million USD, reporting to the Principal Transportation Engineer.
- Compiled RAMS 2023 reporting data for the NHA road network (CEC/PID partnership).
- Conducted Social Impact Assessment surveys for the Khyber Pakhtunkhwa Rural Accessibility Project.
- Performed Road Safety Audit from Shikarpur to D.I. Khan on N-55 (KPRAP).
- Contributed to feasibility studies for RRDP and KPRAP; created Technical Sanctions for four Urban Mobility Projects in Sahiwal and Sialkot.

2024–2025 **Visiting Lecturer**, *University of Engineering and Technology (UET)*, Peshawar, 1 year

- Taught Dynamics and Transportation Engineering-II (Summer 2024); Structural Analysis-I Lab and Building Construction & Drawing.
- Designed course material, assignments, and assessments; supervised lab experiments and guided students through complex engineering concepts.

2020–Present **Freelance Civil Engineer / Programmer**, *Multiple Clients*, 4+ years

- EIDA Automation Framework (C# & MATLAB): seismic vulnerability tool with automated topology extraction, stochastic material assignment, and real-time fragility curves.
- FEM-based software for numerical analysis of ancient masonry structures (MATLAB).
- *Flexpave* and *Conc* desktop applications (C#, WPF, .NET).

2022–2023 **Structural Engineer**, *CEC Minconsult*, 8 months
- Review and redesign of the Women's Business Development Centre, Kohat.
- Foundation design for buildings under the KP Cities Improvement Project; culvert design for RRDP and KPRAP.

2021–2022 **Site Engineer**, *Pakhtunkhwa Highway Authority*, 1 year
- Organised site staff and implemented daily targets in coordination with the Resident Engineer and contractor; reported discrepancies from both ends.

2019–2020 **Trainee Civil Engineer**, *Pakhtunkhwa Highway Authority*, 6 months
- Operated Profilometer, FWD, and HDM-4 for road condition surveys and policy assessment with the Road Asset Management Unit.

Publications

[1] Shafiq, O., Wan Ahmad, S., Ali, S. M., Waqas, M., Khan, G., & Kamarudin, M. A. A. (2025). Fragility Curves Development for Single Span Slab Bridges: A Case Study for Peshawar, Pakistan. *Journal of Advanced Research in Applied Mechanics*, 133(1), 177–190. doi:10.37934/aram.133.1.177190 **[Scopus]**

[2] Syed Mustorpha, S. I., Wan Ahmad, S., Amzar Kamarudin, M. A., Alsakkaf, H. A., & Shafiq, O. (2025). Advancements in Earthquake Research and Seismic Risk Analysis: An In-Depth Review Centered on Malaysia. *Journal of Advanced Research in Applied Sciences and Engineering Technology*, 63(2), 196–206. doi:10.37934/araset.63.2.196206 **[Q2]**

[3] Kamarudin, M. A. A., Wan Ahmad, S., Alsakkaf, H. A., Syed Mustopha, S. I., & Shafiq, O. (2025). The Assessment of Tall Building Structure Performance Due to Seismic Effect in Malaysia. *Journal of Advanced Research Design*, 129(1), 112–129. doi:10.37934/ard.129.1.112129 **[Q4]**

[4] Waqas, M., Wan Ahmad, S., Shafiq, O., Ishaq, F., Laissy, M. Y., Adnan, A., & Al Sadoon, Z. A. (2025). Seismic Fragility Characterization of Grid-Like Frame Structures. *Construction*, 5(2). doi:10.15282/construction.v5i2.13231 **[Q3]**

[5] Omair Shafiq, Abdul Jabbar, Ghufran Ullah. *Achieving High Strength Concrete in Challenging Areas* (2019). Munich: GRIN Verlag. ISBN (eBook): 9783346037336 **[Book]**

[6] Waqas, M., Wan Ahmad, S., Shafiq, O., & Talha, M. (2025). Advances in Earthquake Mitigation in Malaysia: A Review of Global Context and Local Progress. *Jurnal Teknologi*. (Accepted / In Press) **[Q3]**

[7] Shafiq, O., Wan Ahmad, S., Arif, A. I. B. M., Ishaq, F., Khattak, M. M. H., Khan, G., Waqas, M., & Kamarudin, M. A. A. (2025). Factors Affecting Seismic Fragility of RC Grid-Like Frame Structures: A Parametric Meta-Regression Approach. *Structures*. (Revision Submitted — Round 1; MS No. STRUCTURES-D-25-09760) **[Q1]**

[8] Shafiq, O., Wan Ahmad, S., Arif, A. I. B. M., Khan, G., Laissy, M. Y., Faizan, M., & Adnan, A. B. (2026). Probabilistic Seismic Demand Modeling of RC Grid-Like Frames: Assessing Response Sensitivity Across Parametric Structural Configurations. *Construction*. (Under Review) **[Q3]**

[9] Shafiq, O., Wan Ahmad, S., & Iqbal, M. (2025). Parametric Sensitivity of the Seismic Demand Scaling Exponent to Design Variability and Soil–Structure Interaction in Structurally Similar RC Bare Frame Systems. *Structures*. (With Editor) **[Q1]**

[10] Shafiq, O., Wan Ahmad, S., Khan, G., Khattak, M. M. H., & Ishaq, F. (2025). Transparent ML Framework for Seismic Fragility Derivation: A Closed-Form Dimensionless Approach. *Structures*. (Submitted) **[Q1]**

[11] Shafiq, O., Wan Ahmad, S., Jintao, H., & Talha, M. (2025). An Automated Extended Incremental Dynamic Analysis Pipeline for High-Throughput Fragility Data Generation of RC Moment-Frame Buildings with Soil–Structure Interaction in ETABS. *MethodsX*. (Submitted) **[Q1]**

Skills

Programming C# .NET C++ MATLAB Visual Studio

Software ETABS SAP2000 AutoCAD Civil3D HDM-4